

Fenchel-Rockafellar Duality Framework

Start: Primal
 $\min [f(x)+g(x)]$

↑
Compute
Conjugates

↑
Dual Problem
 $\min [f^*(-L^*u)+g^*(u)]$

↑
Check:
0 in $\text{sri}(\dots)$

YES: Strong
Duality

NO: Duality
Gap > 0